

Hydrogen Level Addresses in the UKF Kinematic Framework: A Geometric Rewriting of the Bohr Scale and the $1/n^2$ Spectral Skeleton

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Abstract

Bohr-level radii are represented as dimensionless spatial addresses relative to the proton charge radius. With the address scaling $a_n \propto n^2$ and the kinematic velocity profile $v_n \propto 1/n$, the squared-velocity ratios between levels become exact rational numbers. The characteristic $1/n^2$ structure of the Rydberg formula then appears directly as a difference of squared kinematic velocities:

$$v_i^2 - v_f^2 \propto \frac{1}{n_i^2} - \frac{1}{n_f^2}.$$

A quantitative test shows that applying a single universal reduced-mass correction shifts the spectral residual from -531 ppm to a tight band of $+5.4 \dots +13.2$ ppm across the series. This construction is a geometric rewriting of the Bohr level structure and the combinatorial spectral grid.

1 Parent Scale and Base Speed

In the Unified Kochenov Framework (UKF), physical systems are described via dimensionless spatial addresses scaled to the parent body's characteristic scale. For the hydrogen atom the parent scale is the proton charge radius

$$R_p \approx 0.8412 \times 10^{-15} \text{ m}.$$

Taking the Bohr radius $a_0 \approx 5.29177 \times 10^{-11} \text{ m}$, the base dimensionless address and base kinematic velocity are

$$a_1 = \frac{a_0}{R_p} \approx 6.292 \times 10^4, \quad v_1 = c\alpha \approx 2187.69 \text{ km s}^{-1},$$

where c is the speed of light and $\alpha \approx 1/137.036$ is the fine-structure constant (CODATA).

2 Level Addresses and Characteristic Radial Scales

The spatial address for level n scales as

$$a_n = a_1 n^2.$$

Multiplying back by R_p recovers the classical Bohr radii $r_n = a_0 n^2$.

Physical correspondence of the first five levels:

- $n = 1$: classical Bohr radius a_0 (ground-state scale);
- $n = 2$: characteristic scale of the first excited state;

- $n = 3$: characteristic Bohr radial scale for $n = 3$;
- $n = 4$: sub-nanometre scale typical of simple organic frameworks;
- $n = 5$: threshold of highly excited (Rydberg) atomic states.

Table 1: Hydrogen level addresses, radial scales and kinematic velocities.

Level n	Address a_n	Radial scale r_n	Velocity v_n (km s ⁻¹)
1	6.292×10^4	0.0529 nm	2187.69
2	2.517×10^5	0.2117 nm	1093.85
3	5.663×10^5	0.4763 nm	729.23
4	1.007×10^6	0.8467 nm	546.92
5	1.573×10^6	1.3229 nm	437.54

Extension to hydrogen-like ions

For a single-electron ion with nuclear charge Z

$$r_n(Z) \propto \frac{n^2}{Z}, \quad v_n(Z) \propto \frac{Z}{n}.$$

For He^+ ($Z = 2$) the base velocity doubles and the ground-state radius contracts by a factor of two, recovering the expected Z^2 energy scaling.

3 Squared-Velocity Ratios

From $v^2 \propto 1/a$ and $a \propto n^2$ the ratio of squared velocities between levels n and m is the pure rational

$$\frac{v_n^2}{v_m^2} = \frac{m^2}{n^2}.$$

Table 2: Transition ratios of squared velocities.

Transition $n \leftrightarrow m$	Rational expression	Value
$1 \leftrightarrow 2$	4/1	4.0000
$2 \leftrightarrow 3$	9/4	2.2500
$3 \leftrightarrow 4$	16/9	1.777...
$4 \leftrightarrow 5$	25/16	1.5625

4 Kinematic Skeleton of the Rydberg Difference

The difference of squared kinematic velocities yields

$$v_i^2 - v_f^2 = v_1^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \propto \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right).$$

For a transition $n_i \rightarrow n_f$ with $n_i > n_f$, this quantity is negative, corresponding to photon emission. The wavelength is computed from the absolute value $|\Delta E|$. This is precisely the combinatorial structure of the Rydberg formula. The base kinetic energy recovers the ionisation scale

$$E_1 = \frac{1}{2} m_e v_1^2 \approx 13.60569 \text{ eV}.$$

5 Quantitative Precision and Reduced-Mass Calibration

We evaluate the pure kinematic chain

$$a_n \rightarrow r_n \rightarrow v_n \rightarrow E_n \rightarrow \Delta E \rightarrow \lambda$$

against standard vacuum emission lines.

With the uncorrected factor $E_1 \approx 13.60569$ eV the Lyman- α wavelength is

$$\lambda_{\text{calc}}(2 \rightarrow 1) = 121.50227 \text{ nm.}$$

Relative to the reference value 121.56684 nm the residual is -531.1 ppm.

A single universal reduced-mass factor for hydrogen,

$$\mu_H = \frac{m_p}{m_p + m_e} \approx 0.99945568,$$

rescales the base energy to $E_1^{(\text{red})} = \mu_H E_1$. Re-evaluating the whole series with this one correction produces the results of Table 3.

Table 3: Spectral test: without and with reduced mass.

Transition	without μ	with μ	NIST-Ritz	residual (with μ)
$2 \rightarrow 1$	121.50227 nm	121.56845 nm	121.56684	+13.2 ppm
$3 \rightarrow 2$	656.11228 nm	656.46961 nm	656.46580	+5.8 ppm
$4 \rightarrow 3$	1874.60650 nm	1875.62744 nm	1875.61270	+7.9 ppm
$5 \rightarrow 3$	1281.46929 nm	1282.16720 nm	1282.16030	+5.4 ppm

One and the same physical factor μ moves the entire series from a systematic offset of several hundred ppm into the narrow window $+5.4 \dots +13.2$ ppm. No line-by-line adjustment is performed.

The residual band is consistent with the expected magnitude of unmodelled $\mathcal{O}(\alpha^2)$ fine-structure and QED contributions together with residual metrological uncertainty arising from the adopted values of R_p , α and the reference wavelengths. It is therefore a test residual, not evidence of an additional dynamical field.

6 Methodological Boundaries

Two boundaries are stated explicitly:

1. The macroscopic quadrupole filter J_2 that appears in celestial mechanics scales as $\Delta\theta \propto 1/a \propto n^{-2}$. This does not reproduce the n^{-4} dependence of relativistic fine structure in QED.
2. Absolute frequencies possibly still require the empirical Rydberg constant and the reduced-mass factor. The kinematic construction supplies only the discrete $1/n^2$ combinatorial skeleton.

7 Central Geometric Result

The cleanest statement of the framework remains:

$$a_n \propto n^2 \quad \text{and independently} \quad v_n \propto \frac{1}{n} \quad \implies \quad v_n^2 \propto \frac{1}{n^2} \quad \implies \quad v_i^2 - v_f^2 \propto \frac{1}{n_i^2} - \frac{1}{n_f^2}.$$

This is a geometric reconstruction of the Rydberg combinatorial structure.

8 Proposed Diagnostic Controls

A systematic precision analysis can be organised as the following sequence of pure controls:

1. Address test: $a_n/a_1 \stackrel{?}{=} n^2$;
2. Velocity test: $v_n/v_1 \stackrel{?}{=} 1/n$;
3. Energy test: $E_n/E_1 \stackrel{?}{=} 1/n^2$;
4. Spectral test: $\lambda_{\text{calc}} \rightarrow \lambda_{\text{ref}}$;
5. Uniform reduced-mass test: a single μ applied to the whole series.

After these controls the residual map $\delta_{\text{ppm}}(n_i, n_f, Z)$ may be examined for any systematic dependence on quantum numbers or nuclear charge. Such an examination diagnoses the remaining metrological content.

9 Numerical Summary of the Kinematic Test

All results below are obtained from the pure kinematic chain

$$a_n \propto n^2, \quad v_n \propto 1/n, \quad \Delta(v^2) \propto \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

with a single reduced-mass factor μ and no additional deformation terms.

Table 4: Hydrogen level addresses and kinematic velocities.

n	Address a_n	Radius r_n (nm)	Velocity v_n (km s ⁻¹)
1	6.291×10^4	0.0529	2187.69
2	2.516×10^5	0.2117	1093.85
3	5.662×10^5	0.4763	729.23
4	1.007×10^6	0.8467	546.92
5	1.573×10^6	1.3229	437.54

Table 5: Spectral residuals after a single reduced-mass correction μ .

System	Transition	Calculated (nm)	NIST-Ritz (nm)	Residual
Hydrogen	2 $\rightarrow$ 1	121.56845	121.56684	+13.13 ppm
Hydrogen	3 $\rightarrow$ 2	656.46961	656.46580	+5.80 ppm
Hydrogen	4 $\rightarrow$ 3	1875.62745	1875.61270	+7.86 ppm
Hydrogen	5 $\rightarrow$ 3	1282.16720	1282.16030	+5.38 ppm
Deuterium	2 $\rightarrow$ 1	121.53538	121.53378	+13.13 ppm
Deuterium	3 $\rightarrow$ 2	656.29103	656.28690	+6.29 ppm
Deuterium	4 $\rightarrow$ 3	1875.11723	1875.10110	+8.60 ppm
Lithium-7	3 $\rightarrow$ 2	72.90707	72.90400	+42.05 ppm

The hydrogen and deuterium series fall into a narrow residual band of approximately +5.4 . . . +13.2 ppm.

The larger residual for Li III is expected: the simple one-electron reduced-mass factor does not yet account for higher-order nuclear-size and relativistic corrections that become noticeable at $Z = 3$.

No line-by-line adjustment was applied. The only physical correction used throughout is the universal reduced-mass factor μ .

10 Conclusion

The framework provides a transparent geometric rewriting of the Bohr n^2 grid and of the $1/n^2$ spectral skeleton in purely kinematic variables. Absolute metrology still rests on the Rydberg constant and the reduced-mass correction. The spectral test confirms that a single, physically motivated factor μ brings the entire series into the low-ppm regime, while the methodological boundary stated above is preserved.

11 Full Python Implementation

```
1 # -*- coding: utf-8 -*-
2 """
3 UKF Hydrogenic Spectral Test Kinematic v10.4 (clean)
4 """
5
6 import math
7
8 class UKF_Hydrogenic_Test:
9     def __init__(self):
10         # UKF registers
11         self.M5 = 31
12         self.F12 = 144
13         self.T720 = 720
14         self.phi = (1 + math.sqrt(5)) / 2
15         denom = (self.T720 / 120) * self.M5 - 1 / self.phi
16         self.alpha_inv = 137 + (self.M5**2 / self.F12) / denom
17         self.alpha = 1 / self.alpha_inv
18
19         # Constants
20         self.c = 299792458.0
21         self.hc = 1239.841984 # eV nm
22         self.m_p = 1836.15267343
23         self.m_d = 3670.48296788
24         self.m_li7 = 12786.3927
25
26         # Parent scale
27         self.Rp = 0.8412e-15
28         self.a0 = 5.29177e-11
29         self.a1 = self.a0 / self.Rp
30         self.v1 = self.alpha * self.c / 1000 # km / s
31         self.E1 = 0.5 * 9.1093837e-31 * (self.v1 * 1000)**2 / 1.602176634e-19
32
33     def mu(self, Z, A):
34         if Z == 3 and A == 7:
35             M = self.m_li7
36         elif A == 1:
37             M = self.m_p
38         elif A == 2:
39             M = self.m_d
40         else:
41             raise ValueError(f"Unsupported Z={Z}, A={A}")
42         return M / (M + 1)
43
44     def wavelength(self, Z, A, ni, nf):
45         delta = Z**2 * (1/nf**2 - 1/ni**2)
46         E = self.E1 * self.mu(Z, A) * delta
47         return self.hc / abs(E)
48
49     def run(self):
50         print("=" * 78)
51         print("Table 4. Hydrogen level addresses and kinematic velocities")
52         print("=" * 78)
```

```

53     print(f'{n': <4}{a_n': >14}{r_n (nm)': >14}{v_n (km / s)': >14}")
54     print("-" * 78)
55     for n in range(1, 6):
56         a_n = self.a1 * n**2
57         r_n = a_n * self.Rp * 1e9
58         v_n = self.v1 / n
59         print(f{n: <4}{a_n: >14.3e}{r_n: >14.4f}{v_n: >14.2f}")
60
61     print("\n" + "=" * 90)
62     print("Table 5. Spectral residuals (single reduced - mass correction)")
63     print("=" * 90)
64     print(f'{System': <10}{Trans': <10}{Calc (nm)': >14}{NIST (nm)': >14}{
Residual': >12}")
65     print("-" * 90)
66
67     tests = [
68         ("Hydrogen", 1, 1, 2, 1, 121.56684),
69         ("Hydrogen", 1, 1, 3, 2, 656.46580),
70         ("Hydrogen", 1, 1, 4, 3, 1875.61270),
71         ("Hydrogen", 1, 1, 5, 3, 1282.16030),
72         ("Deuterium", 1, 2, 2, 1, 121.53378),
73         ("Deuterium", 1, 2, 3, 2, 656.28690),
74         ("Deuterium", 1, 2, 4, 3, 1875.10110),
75         ("Li III", 3, 7, 3, 2, 72.90400),
76     ]
77
78     for name, Z, A, ni, nf, ref in tests:
79         calc = self.wavelength(Z, A, ni, nf)
80         ppm = (calc - ref) / ref * 1e6
81         print(f"{name: <10}{ni} -> {nf: <8}{calc: >14.5f}{ref: >14.5f}{ppm:
>+11.2f} ppm")
82
83     print("=" * 90)
84     print(f"= {self.alpha_inv:.8f} | E1 = {self.E1:.6f} eV | v1 = {self.v1:.2f}
km / s")
85
86
87 if __name__ == "__main__":
88     UKF_Hydrogenic_Test().run()

```

Listing 1: UKF Hydrogenic Spectral Test v10.3

Related Works

This paper is part of the Universal Kochenov Framework (UKF) series:

- **Paper 1:** Hydrogen Level Addresses in the UKF Kinematic Framework: A Geometric Rewriting of the Bohr Scale and the $1/n^2$ Spectral Skeleton
10.5281/zenodo.21940531
- **Paper 2:** Fractal Cascade: From Proton to Solar Photosphere
10.5281/zenodo.21940552
- **Paper 3:** A Mass-Free Kinematic Topology for Orbital Mechanics
10.5281/zenodo.21940519

All papers are available as preprints on Zenodo under the UKF series.